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United States Patent	12411971
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Kwong; Helen et al.

Authenticating accesses to a shared datastore of a multi-tenant computer system

Abstract

A multi-tenant computer system authenticates access to a shared datastore by a shared service running on the multi-tenant computer system. The shared service is operable to access the shared datastore to execute requests from a plurality of multi-tenant cloud computing services. The requests include an indication of a particular tenant and a particular tenant grouping indicator. Requests are authenticated by cryptographically verifying the request and verifying that the particular tenant is associated with the particular tenant grouping indicator. In response to authenticating a request, the shared service accesses the shared datastore to execute the first request.

Inventors:	Kwong; Helen (Danville, CA), Busjaeger; Benjamin (San Carlos, CA)
Applicant:	Salesforce, Inc. (San Francisco, CA)
Family ID:	84285118
Assignee:	Salesforce, Inc. (San Francisco, CA)
Appl. No.:	18/482407
Filed:	October 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240037267 A1	Feb. 01, 2024

Related U.S. Application Data

continuation parent-doc US 17338437 20210603 US 11783071 child-doc US 18482407

Publication Classification

Int. Cl.: **G06F21/62** (20130101); **G06F21/60** (20130101); **H04L9/40** (20220101)

U.S. Cl.:

CPC **G06F21/6218** (20130101); **G06F21/602** (20130101); **G06F21/604** (20130101);
 H04L63/104 (20130101);

Field of Classification Search

USPC: None

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Primary Examiner: Perungavoor; Venkat

Attorney, Agent or Firm: Kowert, Hood, Munyon, Rankin & Goetzel, P.C.

Background/Summary

PRIORITY CLAIM (1) The present application is a continuation of U.S. application Ser. No. 17/338,437, entitled “AUTHENTICATING ACCESSES TO A SHARED DATASTORE OF A MULTI-TENANT COMPUTER SYSTEM,” filed Jun. 3, 2021, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

Technical Field

(1) This disclosure relates generally to multi-tenant computer systems.

Description of the Related Art

(2) Software-level multi-tenancy enables a single instance of software to run on computer hardware and provide services to a plurality of users. With multi-tenancy, a software application is designed to provide tenants with a share of the instance such that data, metadata, configurations, settings, user management and other functions are isolated from the other tenants. Tenants may store information in a shared datastore of the multi-tenant computer system. Moreover, the tenants of a multi-tenant computer system may themselves provide services to their own tenants. These subtenants may also store information in a shared datastore of the multi-tenant computer system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating an embodiment of a computer system configured to authenticate and execute requests made by multi-tenant cloud computing services to shared services.

(2) FIG. 2 is an expanded block diagram of the multi-tenant computer system of FIG. 1 in accordance with various embodiments.

(3) FIG. 3 is a block diagram illustrating authentication and execution of a request made to shared service in accordance with various embodiments.

(4) FIG. 4 is block diagram of a data protection scope hierarchy useable to determining whether requests are authorized in accordance with various embodiments.

(5) FIG. 5 is an example token sent from a multi-tenant cloud computing service to a shared service in accordance with various embodiments.

(6) FIG. 6 is an example token sent from a shared service to a shared datastore in accordance with various embodiments.

(7) FIG. 7 is flowchart illustrating an embodiment of an authentication and execution method in accordance with the various embodiments.

(8) FIG. 8 is a block diagram of an exemplary computer system, which may implement the various components of FIGS. 1, 2, and 3.

(9) This disclosure includes references to “one embodiment” or “an embodiment.” The appearances of the phrases “in one embodiment” or “in an embodiment” do not necessarily refer to the same embodiment. Particular features, structures, or characteristics may be combined in any suitable manner consistent with this disclosure.

(10) Within this disclosure, different entities (which may variously be referred to as “units,” “circuits,” other components, etc.) may be described or claimed as “configured” to perform one or more tasks or operations. This formulation—[entity] configured to [perform one or more tasks]—is used herein to refer to structure (i.e., something physical, such as an electronic circuit). More specifically, this formulation is used to indicate that this structure is arranged to perform the one or more tasks during operation. A structure can be said to be “configured to” perform some task even if the structure is not currently being operated. A “computer system configured to authenticate requests” is intended to cover, for example, a computer system has circuitry that performs this function during operation, even if the computer system in question is not currently being used (e.g., a power supply is not connected to it). Thus, an entity described or recited as “configured to” perform some task refers to something physical, such as a device, circuit, memory storing program instructions executable to implement the task, etc. This phrase is not used herein to refer to something intangible. Thus, the “configured to” construct is not used herein to refer to a software

entity such as an application programming interface (API).

(11) The term “configured to” is not intended to mean “configurable to.” An unprogrammed FPGA, for example, would not be considered to be “configured to” perform some specific function, although it may be “configurable to” perform that function and may be “configured to” perform the function after programming.

(12) Reciting in the appended claims that a structure is “configured to” perform one or more tasks is expressly intended not to invoke 35 U.S.C. § 112(f) for that claim element. Accordingly, none of the claims in this application as filed are intended to be interpreted as having means-plus-function elements. Should Applicant wish to invoke Section 112(f) during prosecution, it will recite claim elements using the “means for” [performing a function] construct.

(13) As used herein, the terms “first,” “second,” etc. are used as labels for nouns that they precede, and do not imply any type of ordering (e.g., spatial, temporal, logical, etc.) unless specifically stated. For example, references to “first” and “second” multi-tenant cloud computing services would not imply an ordering between the two unless otherwise stated.

(14) As used herein, the term “based on” is used to describe one or more factors that affect a determination. This term does not foreclose the possibility that additional factors may affect a determination. That is, a determination may be solely based on specified factors or based on the specified factors as well as other, unspecified factors. Consider the phrase “determine A based on B.” This phrase specifies that B is a factor is used to determine A or that affects the determination of A. This phrase does not foreclose that the determination of A may also be based on some other factor, such as C. This phrase is also intended to cover an embodiment in which A is determined based solely on B. As used herein, the phrase “based on” is thus synonymous with the phrase “based at least in part on.”

(15) “In this disclosure, various “modules” operable to perform designated functions are shown in the figures and described in detail (e.g., tenant group module **150**, etc.). As used herein, a “module” refers to software or hardware that is operable to perform a specified set of operations. A module may refer to a set of software instructions that are executable by a computer system to perform the set of operations. A module may also refer to hardware that is configured to perform the set of operations. A hardware module may constitute general-purpose hardware as well as a non-transitory computer-readable medium that stores program instructions, or specialized hardware such as a customized ASIC. Accordingly, a module that is described as being “executable” to perform operations refers to a software module, while a module that is described as being “configured” to perform operations refers to a hardware module. A module that is described as “operable” to perform operations refers to a software module, a hardware module, or some combination thereof. Further, for any discussion herein that refers to a module that is “executable” to perform certain operations, it is to be understood that those operations may be implemented, in other embodiments, by a hardware module “configured” to perform the operations, and vice versa.”

DETAILED DESCRIPTION

(16) Outside of a multi-tenant computer system, some tenants and subtenants may be associated with each other and desire to share access to information stored in a shared datastore of the multi-tenant computer system. For example, if Company ABC is a tenant of two different multi-tenant cloud computing services (also referred to herein as “subtenancy”), Company ABC may desire that both tenants be able to use a shared service to access information stored in the shared datastore. In a multi-tenant computer system that facilitates multiple multi-tenant cloud services that themselves have their own tenants, sharing access to information stored in shared datastores can be complex to implement. In particular, because tenancy can be defined in any of a number of ways and authenticated using any number of techniques, just because the same multi-tenant computer system implements two multi-tenant cloud services does not mean there are no barriers between the two multi-tenant cloud services. Thus, the credentials usable to access a first multi-tenant cloud service may not be useable to access a second multi-tenant cloud service. Further, to maintain information

security, access to information in shared datastores is restricted (e.g., only the tenant that stored information may access it, only certain tenants may access the shared datastore).

(17) As a non-limiting example, Company ABC is a shoe company. Company ABC uses a multi-tenant computer system that implements a first multi-tenant cloud computing service that is a customer relationship management service that maintains contacts with buyers of shoes, a second multi-tenant cloud computing service that is a commerce service that allows customers to purchase shoes via Company ABC's website, and a shared inventory service that is usable by the multi-tenant cloud computing services to access and maintain information about the number of shoes in inventory. Company ABC may desire that its two tenants have access to the number of shoes in inventory. For example, Company ABC may want the customer relationship management service to have access to the number of shoes in stock to drive interactions with buyers at shoe stores and the commerce service to have access to the same information to allow customers to make direct purchases on Company ABC's website. To do so, Company ABC would need for both multi-tenant cloud computing services to be able to use the inventory service to access the information in the shared datastore. In order to maintain information security, the multi-tenant computer system ensures that both tenants are authorized to access the inventory information by determining whether there is a trust relationship between the tenant that owns the information and other Company ABC tenants.

(18) Establishing trust relationships one-at-a-time (e.g., Tenant One extends trust to Tenant Two) can result in a cumbersome network of trust relations that consumes computational resources to manage and maintain. This one-at-a-time approach of “extending trust” may become unwieldy at least because as the number of subtenants, multi-tenant cloud computing services, and shared services increases that size of the network of extended trust relationships increases. Additionally, if an entity discontinues use of the first multi-tenant cloud computing service, this extended trust may have to be reworked such that Tenant Two becomes the owner of the shared information or the discontinued account for subtenant one merely for the purposes of maintaining the extended trust relationship.

(19) Rather than establishing trust relationships one-at-a-time from one tenant to another tenant, the techniques proposed herein use “tenant groups” to establish trust relationships. Instead of determining, for example, whether the tenant that owns information stored in the shared datastore has extended trust to another tenant, access to that information may be granted to a particular tenant group. Membership in the tenant group may be used to determine whether a tenant is authorized to access information. In order to facilitate the use of tenant groups to authenticate information accesses, the inventors here have developed the following techniques of creating tenant groups and adding tenants to the tenant groups, invoking tenant group membership in requests to access the information stored in the shared datastore, and authenticating access requests by (a) using cryptographic techniques to verify the authenticity of requests and (b) determining whether the tenant making the request is a member of the tenant group.

(20) FIG. 1 is a block diagram illustrating an embodiment of a computer system **100** including a multi-tenant computer system **120** configured to authenticate and execute requests made by multi-tenant cloud computing services **122** to shared services **124** using tenant groupings.

(21) As shown in FIG. 1, multi-tenant computer system **120** communicates with one or more remote computer systems **110** (e.g., first remote computer system **110A**, second remote computer system **110**). In various embodiments, such communication includes communication over one or more local area networks and/or wide area networks (e.g., the Internet). In various embodiments, remote computer systems **110** are used by users to access multi-tenant computer system **120** and send requests to components thereof. In various instances, such requests include requests made to multi-tenant cloud computing services **122** that result in requests being sent to shared services **124** for execution as discussed herein. While two remote computer systems **110** are shown in FIG. 1 (i.e., first remote computer system **110A** and second remote computer system **110B**), it will be

understood that any number of remote computer systems **110** may be present (e.g., tens, hundreds, thousands, or millions of remote computer systems **110**). In various embodiments, individual remote computer systems **110** are implemented by software running on a computer system (e.g., a desktop computer, a laptop computer, a tablet computer, a mobile phone, a server, a wearable computer) or a plurality of computer systems (e.g., a network of servers operating as a cloud). In other embodiments, individual remote computer systems **110** are implemented in specialized hardware (e.g., on an FPGA) or in a combination of hardware and software.

(22) In various embodiments, multi-tenant computer system **120** is implemented by software running on a computer system (e.g., a desktop computer, a laptop computer, a tablet computer, a mobile phone, a server) or a plurality of computer systems (e.g., a network of servers operating as a cloud). In other embodiments, multi-tenant computer system **120** is implemented in specialized hardware (e.g., on an FPGA) or in a combination of hardware and software. In various embodiments, multi-tenant computer system **120** implements one or more multi-tenant cloud computing services **122**, one or more shared services **124**, one or more shared datastores **130** as well as a cloud-to-cloud interface service **140** and a tenant group module **150**. In the embodiment shown in FIG. **1**, multi-tenant computer system **120** implements a first multi-tenant cloud computing service **122A**, a second multi-tenant cloud computing service **122B**, a first shared service **124A**, a second shared service **124B**, and a single shared datastore **130**, but in other embodiments, multi-tenant computer system **120** implements any number of these components. Additionally, tenant group module **150** and cloud-to-cloud interface service **140** may be implemented separately as shown in FIG. **1** or in a single module. Moreover, the operations performed by the various components of multi-tenant computer system **120** may be distributed across multiple instances of the components (e.g., multiple instances of first shared service **124A** that share the load).

(23) In various embodiments, multi-tenant cloud computing services **122** (including first multi-tenant cloud computing service **122A** and a second multi-tenant cloud computing service **122B**) are tenants of multi-tenant computer system **120** and are instances of software that can be accessed by multiple subtenants according to a multi-tenancy model and provide functions to their subtenants. As used herein, “multi-tenant cloud computing services” refers to software that is accessible to its tenants and is useable to perform functions for such tenants (this is distinct from a cloud environment like AMAZON WEB SERVICES that provides distributed hardware and software on which the cloud environment's users can run software). As discussed in further detail in reference to FIG. **2**, different multi-tenant cloud computing services **122** may have different sets of tenants and define tenancy differently. Thus, the respective multi-tenant cloud computing services **122** have their own respective sets of tenants (also referred to herein as “subtenants”) while the various multi-tenant cloud computing services **122** are themselves tenants of multi-tenant computer system **120**. In various embodiments, first multi-tenant cloud computing service **122A** performs different functions for its subtenants than second multi-tenant cloud computing service **122B** performs for its subtenants (e.g., first multi-tenant cloud computing service **122A** may perform various tasks relating to customer relationship management such as storing contact information of customers while second multi-tenant cloud computing service **122B** may perform various tasks relating to fulfilling orders from customers).

(24) As discussed below, the multi-tenant cloud computing services **122** are operable to send requests, on behalf of their subtenants, to one or more shared services **124**, the execution of which include accessing shared datastore **130**. Shared services **124** (including first shared service **124A** and a second shared service **124B**) are services running on multi-tenant computer system **120** that are operable to access shared datastore **130** to execute requests sent from the various multi-tenant cloud computing services **122**. As used herein, a “shared service” is software that is operable to access, change, and/or manage data in shared datastore **130** on behalf of multi-tenant cloud computing services **122** and/or tenants of multi-tenant cloud computing services **122**. In contrast to

multi-tenant cloud computing services **122**, in various embodiments shared services **124** do not have their own tenants but instead execute tasks sent by multi-tenant cloud computing services **122** on behalf of tenants of the multi-tenant cloud computing services **122**. In such embodiments, while shared services **124** may keep track of which tenants (or tenant groups) of multi-tenant cloud computing services **122** use the shared services **124** and which tenants have been provisioned to use shared datastore **130**, such tenants nevertheless belong to multi-tenant cloud computing services **122**. In other embodiments, however, shared services **124** may have their own tenants. As defined herein, a shared service **124** that has its own tenants but is operable to access, change, and/or manage data on behalf of multi-tenant cloud computing services **122** is still referred to herein as a “shared service” and is not a “multi-tenant cloud computing service” as defined herein.

(25) In various embodiments, first shared service **124A** is operable to perform different functions for multi-tenant cloud computing services **122** than second shared service **124B** (e.g., first shared service **124A** is operable to access and manage information stored in shared datastore **130** relating to inventories of goods and second shared service **124** is operable to access and manage information stored in shared datastore **130** relating to order fulfillment). In various embodiments, one or more of the shared services **124** are also operable to execute requests sent by tenants of multi-tenant computer system **120** that are not themselves multi-tenant cloud computing services **122**. For example, such a tenant may be a user account directly registered with multi-tenant computer system **120** (i.e., a user account that is not a subtenant of a multi-tenant cloud computing services **122**). Such a user account is referred to herein as a “direct tenant” of multi-tenant computer system **120**. Additionally, as discussed above, in various instances, a shared service **124** may have its own tenants. Thus, a given shared service **124** may be operable to access, change, and/or manage data in shared datastore **130** on behalf of multi-tenant cloud computing services **122**, direct tenants of multi-tenant computer system **120**, and/or tenants of the given shared service **124** in various embodiments.

(26) As discussed herein, while the various multi-tenant cloud computing services **122** may have respective sets of tenants with tenancy defined in different ways, in various instances tenants for one multi-tenant cloud computing service **122** (e.g., first multi-tenant cloud computing service **122A**) correspond with the same real-world entities as tenants for another multi-tenant cloud computing service **122** (e.g., second multi-tenant cloud computing service **122B**). For example, Company ABC may have three tenants of first multi-tenant cloud computing service **122A** and four tenants of second multi-tenant cloud computing service **122B**. As discussed below, in various embodiments, some or all of these seven tenants corresponding to Company ABC may be associated together in a tenant group (e.g., in Tenant Group A **200** shown on FIG. 2). The multi-tenant cloud computing services **122**, shared services **124**, and shared datastore **130** are discussed in further detail herein in references to FIG. 2.

(27) In various embodiments, multi-tenant computer system **120** implements a tenant group module **150** that is operable to store information (a) relating to the various tenant groups including but not limited to unique identifiers for the various tenant groups and (b) relating to which tenants of multi-tenant computer system **120** and/or subtenants of the various multi-tenant cloud computing services **122** are associated with a particular tenant group. As discussed herein, in various embodiments, the members of a tenant group are associated with a common entity (e.g., a particular person or organization). While the phrase “tenant group” is used herein, it will be understood that in various embodiments, a “tenant group” may include direct tenants of multi-tenant computer system **120** and/or subtenants.

(28) As discussed in further detail in reference to FIGS. 2-6, the membership of a particular tenant group defines a trust relationship such that, with regard to the information stored in shared datastore **130** that is accessed by one or more shared services **124**, the level of access that a particular tenant of the tenant group is equal to the level of access of other tenants in the tenant group. For example, shared datastore **130** may store values that are indicative of the amount of a

particular good that is stored in a warehouse (e.g., a particular size of a particular type of shoe). As discussed herein, however, a particular shared service **124** may access and/or revise these values as part of executing a request from different multi-tenant cloud computing services **122**. In some embodiments, for example, first shared service **124** may access and/or revise these values to reflect incoming shipments from a supplier (as part of executing a request from first multi-tenant cloud computing service **122A**) or outgoing shipments from the warehouse to a distribution center (as part of executing a request from second multi-tenant cloud computing service **122B**). Thus, different tenants of different multi-tenant cloud computing service **122** are able to make requests that involve accessing these values in the shared datastore **130** (e.g., to read, write, or both) via a particular shared service **124**.

(29) In some embodiments, information in shared datastore **130** is stored such that information stored using a first shared service **124A** can be made accessible to tenants of the various multi-tenant cloud computing services **122** only via first shared service **124A**, but such information is not accessible to second shared service **124B** (even if a tenant using second shared service **124** might have access to the information via first shared service **124A**). In various embodiments, for example, first shared service **124A** and second shared service **124B** may have incompatible data models (e.g., first shared service **124A** is an inventory service and second shared service **124B** is a customer data service that manages personal information for customers such as the customers' ages and contact information). In other embodiments, however, information may be shared across different shared services **124**. For example, in an embodiment in which first shared service **124A** is an inventory service that maintains inventory values, second shared service **124B** may be an electronic shopping chart service that is operable to access and/or revise these values. The electronic shopping cart service may be operable to determine whether the particular good is available for purchase (as part of executing a request from first multi-tenant cloud computing service **122A**) and place holds on inventory during customer checkout and remove holds after checkout (as part of executing a request from second multi-tenant cloud computing service **122B**).

(30) In order to maintain informational security, multi-tenant computer system **120** is operable to authenticate requests to shared services **124** (e.g., first shared service **124A**, second shared service **124B**). In various embodiments, such requests include (a) an indication of the tenant that is making the request (e.g., a particular tenant of first multi-tenant cloud computing service **122A**) and (b) a tenant grouping indicator. As discussed in further detail in reference to FIGS. 2 and 3, a tenant grouping indicator may be any suitable identifier of a tenant group including a universally unique identifier, a name of the tenant group, etc. In various embodiments, multi-tenant computer system **120** determines that a particular request made to shared service **124** is authentic by (a) cryptographically verifying the particular request and (b) verifying that the tenant associated with the request is associated with the tenant group indicated in the request. In various embodiments, cloud-to-cloud interface service **140** is used to cryptographically authenticate the request (e.g., by performing one or more cryptographic operations to determine whether the appropriate cryptographic key was used to sign the request). In various embodiments, tenant group module **150** is used to determine whether a particular tenant is in fact a member of the tenant group indicated in the request. In various embodiments, tenant group module **150** includes a lookup table that determines whether a particular tenant group (identified by a tenant group indicator) includes the tenant that made the request.

(31) If the request is authenticated, the shared service **124** is operable to execute the request including accessing shared datastore **130**. In various embodiments, shared datastore **130** is also operable to authenticate the request in a similar manner by (a) cryptographically verifying the particular request, and (b) verifying that the tenant associated with the request is associated with the tenant group indicated in the request. In various embodiments, shared datastore **130** is also operable to determine whether the tenant group has access to the requested resource (e.g., a directory in shared datastore **130**). While cloud-to-cloud interface service **140** and tenant group module **150** are

represented as separate components in FIG. 1, in various embodiments cloud-to-cloud interface service **140** and tenant group module **150** may be implemented separately or in a single component. Cloud-to-cloud interface service **140** and tenant group module **150** are discussed in further detail in reference to FIG. 3.

(32) Accordingly, multi-tenant computer system **120** is operable to facilitate requests made to a particular shared service **124** by tenants of various multi-tenant cloud computing services **122** and facilitate access by the particular shared service **124** to a shared datastore **130** to execute the request. The information stored in shared datastore **130** is secured by authenticating the request (a) cryptographically to ensure that the requestor (e.g., the requesting subtenant of a multi-tenant cloud computing service **122**) has presented authentic credentials and (b) by verifying that the requestor is a member of the particular tenant group with access to that information stored in shared datastore **130**. Because access to this information is verified by determining whether the requestor is associated with the particular tenant group, determining whether a particular requestor is trusted to access the information is based on association with the particular tenant group itself and not in relationship to other tenants that are associated with the tenant group.

(33) In contrast to determining whether a requestor is authorized using tenant group membership, authorization might be done piecemeal. For example, subtenant one of first multi-tenant cloud computing service **122A** may write a value to shared datastore **130** via first shared service **124A** and authorize subtenant two of second multi-tenant cloud computing service **122B** to access the value in shared datastore **130** via first shared service **124A**. This one-at-a-time approach of “extending trust” may become unwieldy, however, because as the number of subtenants, multi-tenant cloud computing services **122**, and shared services **124** increases the size of the network of extended trust relationships increases. For example, in an embodiment in which three multi-tenant cloud computing services **122** are present and configured to access shared datastore **130** via three shared services **124**, six extended trust relationships are needed to allow access to the same information stored in shared datastore **130** to subtenants of each multi-tenant cloud computing service **122** acting via each shared service. If four multi-tenant cloud computing services **122** and four shared service **124** are present, this number increases to twelve. Further, if a single entity owns more than one subtenant in multi-tenant cloud computing services **122**, this number increases further still. Additionally, if an entity discontinues use of first multi-tenant cloud computing service **122A**, this extended trust may have to be reworked such that subtenant two becomes the owner of the value or the discontinued account for subtenant one merely for the purposes of maintaining the extended trust relationship.

(34) Through the use of tenant groups, however, multi-tenant computer system **120** is operable to restrict access to information based on membership in a tenant group: access is granted based on membership in the tenant group with nonmembers being denied access. An entity can enroll its various tenants in a particular tenant group, and the particular tenant group can be used as the entity's identity across the various multi-tenant cloud computing services **122** and shared services **124**. This approach is able to scale as the number of multi-tenant cloud computing services **122**, shared services **124**, and tenants/subtenants increases. In contrast to an increasing number of trust relationships extended one-at-a-time, the entity's identity across the multi-tenant computer system **120** is maintained by membership in the entity's tenant group. Further, rather than each shared service **124** separately maintaining lists of trust relationships, the shared services **124** can verify that access to information is authorized by querying tenant group module **150** to determine whether a requesting subtenant is a member of the claimed tenant group. If a subtenant account is removed (e.g., because the entity has discontinued use of first multi-tenant cloud computing service **122A**), removing the subtenant account from the tenant group will not require reprovisioning a network of extended trust relationships. If a new multi-tenant cloud computing services **122** is added to multi-tenant computer system **120**, subtenants for the new multi-tenant cloud computing services **122** can be added to the tenant groups to grant access to shared datastore **130**. If a new shared service **124** is

added to is added to multi-tenant computer system **120**, the shared service **124** can use the existing tenant groups to verify access requests. If an entity changes (e.g., through a merger, acquisition, reorganization), the entity can revise its identity within multi-tenant computer system **120** by changing the membership of its tenant group or adding/removing tenant groups. Accordingly, through the use of tenant groups, facilitating information security in shared datastore **130** is made easier and in a more scalable fashion than through the use of tenant-to-tenant extended trust relationships.

(35) FIG. 2 is an expanded block diagram of the multi-tenant computer system **120** in accordance with various embodiments. FIG. 2 depicts first multi-tenant cloud computing service **122A**, second multi-tenant cloud computing service **122B**, first shared service **124A**, second shared service **124B**, shared datastore **130**, and tenant group module **150** in additional detail. As shown in FIG. 2, tenants of multi-tenant cloud computing services **122A** and **122B** (which are subtenants of multi-tenant computer system **120** as discussed herein) are grouped together in tenant groups. First multi-tenant cloud computing service **122A** and second multi-tenant cloud computing service **122B** are operable to send requests to first shared service **124A** and second shared service **124B**. First shared service **124A** and second shared service **124B** are operable to access shared datastore **130** to execute such requests and to report results of the execution of the requests to first multi-tenant cloud computing service **122A** and second multi-tenant cloud computing service **122B**. As shown in FIG. 2 and discussed below, in various embodiments, logical separations are implemented by putting contexts (in the form of URL paths) around both the requests to the shared services **124A** and **124B** and the execution of the requests to access particular directories or subdirectories in shared datastore **130**. Alternatively, rather than dividing information into directories or subdirectories via a file system, information may be stored in shared datastore **130** in the form of a shared table that is accessible by a database service that controls accesses to shared datastore **130** such that information owned by a first tenant group is not accessible to other tenant groups. In such embodiments, the context put around requests may be used to indicate which portions of the shared table (e.g., which row or column) the request is attempting to access.

(36) In the embodiment shown in FIG. 1, first multi-tenant cloud computing service **122A** has two tenants: tenant **A01 202** and tenant **A02 204** and second multi-tenant cloud computing service **122B** similarly has two tenants: tenant **B01 212** and tenant **B02 214**. In the embodiment shown in FIG. 1, two tenant groups are depicted: tenant group **A 200** and tenant group **B 210**. Tenant group **A 200** includes tenant **A01 202** and tenant **B01 204**. Tenant group **B 210** includes tenant **A02 212** and tenant **B02 214**. It will be understood that having four tenants and two tenant groups is merely an example. Any number of tenants or tenant groups may be present in various embodiments.

(37) As discussed above, in various embodiments, tenancy in first multi-tenant cloud computing service **122A** and second multi-tenant cloud computing service **122B** may be defined differently. As a result, for example, the credentials (e.g., username, password, tokens, etc.) used to authenticate tenants of each may be different. The APIs exposed by first multi-tenant cloud computing service **122A** and second multi-tenant cloud computing service **122B** to remote computer systems may be different and be called using different parameters. The access URLs of first multi-tenant cloud computing service **122A** and second multi-tenant cloud computing service **122B** may differ. In short, while an entity may control one or more tenants at both first multi-tenant cloud computing service **122A** and second multi-tenant cloud computing service **122B**, these tenants are not interoperable between first multi-tenant cloud computing service **122A** and second multi-tenant cloud computing service **122B**.

(38) Tenant groups like tenant group **A 200** and tenant group **B** may be implemented using lists or tables in which members of the group are recorded and associated with one or more identifiers of the tenant group. As discussed herein, the tenant group can serve as both an alias for members (e.g., a request sent on behalf of tenant **A01 202** includes an indicator of tenant group **A 200** to execute the request using the credentials for tenant group **A 200** and not tenant **A01 202** individually) and as

a way to establish trust relationships (e.g., when a member is added to the tenant group, the member is granted access to resources to which the tenant group has been granted access). In various embodiments, tenant group module **150** records an indication (e.g., a public key, a unique identifier) corresponding to the various members of each tenant group. When an indication of a tenant group is presented along with an indication of a tenant that is requesting access, tenant group module **150** is operable to be queried as discussed herein to use the tenant group indicator to look up a particular tenant group and determine whether the indication of the tenant is listed as being a member.

(39) In various embodiments, tenant groups have “owning tenants” that control membership of the tenant group and are authorized to add or remove members. If an owning tenant is a member of the tenant group, it is referred to herein as a “primary member” of the tenant group, but in other embodiments an owning tenant is not a member of the tenant group it owns. In various embodiments, tenant groups only have one owning tenant, but in other embodiments membership in a tenant group can be implemented using tiers such that members on the primary tier can add or remove members while members below the primary tier cannot add or remove members. For example, the primary member of tenant group A is tenant **A01 202**. As discussed in additional detail in reference to FIGS. **3**, **5**, and **6**, the level of access given to the members of the tenant group is equal to the level access given to the primary member. Thus, if tenant **A01 202** has access to a particular directory in shared datastore **130**, tenant **B01 204** also has access to that particular directory. In embodiments in which the owning tenant is not a member of the tenant group it owns, while the owning tenant can control membership of the tenant group, the owning tenant cannot invoke the tenant group in a request to a shared service **124** to access information in shared datastore **130** as discussed herein.

(40) In various embodiments, tenant group module **150** stores one or more indicators of the various tenant groups (e.g., tenant group A **200**, tenant group B **210**). In various embodiments, for a given tenant group, tenant group module **150** stores a universally unique identifier (UUID) that has been assigned to the given tenant group when it was created (e.g., when the tenant group was initially provisioned and given at least one member). In various embodiments, the UUID for a tenant group is a label (e.g., a 128-bit or 256-bit code). In some embodiments, the UUID is generated by tenant group module **150**, but in other embodiments is generated by cloud-to-cloud interface service **140**. Additionally or alternatively, in various embodiments, for a given tenant group, tenant group module **150** stores a “natural key” generated from a name of the tenant group (e.g., a name input by a user, a name generated by multi-tenant computer system **120**) and a key corresponding to the primary member (or one of the primary members if there is more than one). This key may be, for example, a public key that is assigned to the primary member by cloud-to-cloud interface service **140** or even by the multi-tenant cloud computing service to which the primary member is a tenant. In various embodiments, however, because the primary member of a tenant group may be changed to a different member, while unique like the UUID, the natural key may change. Thus, in various embodiments, a tenant group may be identified by a stable UUID or by a mutable natural key. As used herein, a “tenant grouping indicator” includes but is not limited to a UUID or natural key corresponding to a tenant group.

(41) As discussed herein, first multi-tenant cloud computing service **122A** and second multi-tenant cloud computing service **122B** are operable to send requests to first shared service **124A**, the execution of which include accessing shared datastore **130** and to send requests to second shared service **124B**, the execution of which also includes accessing shared datastore **130**. In various embodiments, sending such a request includes appending an identifier of the shared service (e.g., /1stservice for first shared service **124A**, /2ndservice for second shared service **124B**) to a URL to which the address is sent. In so doing, logical separation can be enforced between requests sent to first shared service **124A** and second shared service **124B**.

(42) Subsequent to receiving a request from first multi-tenant cloud computing service **122A** or

second multi-tenant cloud computing service **122B**, the shared services **124** are operable to authenticate the request in various embodiments. As discussed herein, authenticating the request may include (a) cryptographically verifying the particular request and (b) verifying that the tenant associated with the request is associated with the tenant group indicated in the request. As discussed herein, through the use of tenant groups, different shared services **124** are operable to authenticate requests based on the requesting tenant's membership in a tenant group. Thus, both first shared service **124A** and second shared service **124B** are operable to use the same information (e.g., an indication of a particular tenant group and a list of members of the particular tenant group) to authenticate requests and without having to maintain separate networks of extended trust relationships. In various embodiments, tenant groupings are the only representation of trust relationships between tenants of the plurality of multi-tenant cloud computing services **122** (e.g., first multi-tenant cloud computing services **122A** and second multi-tenant cloud computing services **122B**) used by shared services **124** (e.g., first shared service **124A** and second shared service **124B**) to authenticate requests. Authentication of requests by shared services is discussed in additional detail in reference to FIGS. **3** and **5**.

(43) Subsequent to authenticating the requests, shared services **124** are operable to access shared datastore **130** to execute the requests. When the shared services **124** access shared datastore **130**, the shared service accesses the shared datastore **130** using a URL that provides the context of the shared service executing the request and the context of the tenant group in the request. For example, when first shared service **124A** accesses shared datastore **130**, URL suffix **240** is used in various embodiments which indicates that a particular access is done in the context of services generally, first shared service **124A** in particular, tenant groups generally, and a particular tenant group (e.g., tenant group A **200**) in particular. Similarly, in various embodiments, when second shared service **124B** accesses shared datastore **130**, URL suffix **242** is used which indicates that a particular access is done in the context of services generally, second shared service **124B** in particular, tenant groups generally, and a particular tenant group (e.g., tenant group A **200**) in particular.

(44) In various embodiments, shared services **124** are operable to access particular directories and subdirectories of shared datastore **130** as part of executing the requests. In various embodiments, shared datastore **130** includes directories (or portions of a shared table) corresponding to the various shared services **124** in multi-tenant computer system **120** and subdirectories corresponding to various tenant groups. Accordingly, in the example depicted in FIG. **2**, shared datastore **130** includes a directory A **220** (corresponding to first shared service **124A** and located at /1stservice) and a directory B **230** (corresponding to second shared service **124B** and located at /2ndservice). Directory A **220** includes a subdirectory **222A** (located at /1stservice/groupA) corresponding to tenant group A **200** and a subdirectory **222B** (located at /1stservice/groupB) corresponding to tenant group B **210**. Directory B **230** includes a subdirectory **232A** (located at /2ndservice/groupA) corresponding to tenant group A **200** and a subdirectory **232B** (located at /2ndservice/groupB) corresponding to tenant group B **210**. While shared datastore **130** is discussed herein as being implemented using directories containing subdirectories or a shared table divided into rows and columns, it will be understood these are merely examples and that any of a number of techniques to logically separate information in shared datastore **130** may be used. Shared services **124** accesses to shared datastore **130** are discussed in additional detail in reference to FIGS. **3** and **6**.

(45) In various embodiments, to maintain informational security, shared services **124** are only operable to access their corresponding directory in shared datastore **130** and shared datastore **130** is operable to only permit shared services **124** to access their corresponding directories. Thus, first shared service **124A** is operable only to access directory A **220** in shared datastore **130** and second shared service **124B** is operable only to access directory B **230** in shared datastore **130**. Moreover, shared datastore **130** is operable to prevent first shared service **124A** from accessing directory B **230** and to prevent second shared service **124B** from accessing directory A **220**. Similarly, accesses

to shared datastore **130** on behalf of particular tenants requesting access in the context of a tenant group is restricted to just the subdirectory corresponding to that tenant group. Thus, tenants **A01 202** and **B01 204** acting in the context of tenant group **A 200**, are able to access information in subdirectory **222A** via requests to first shared service **124A**. Thus, tenant **A01 202** may use shared service **124A** to update a value stored in subdirectory **222A** and tenant **B01 212** may use shared service **124A** to access that value. However, because subdirectory **222A** is logically isolated from other directories, tenants **A02 204** and **B02 214** are not able to access the value. Similarly, tenants **A01 202** and **B01 204** acting in the context of tenant group **A 200**, are able to access information in subdirectory **222B** via requests to first shared service **124B**. In some embodiments, values in subdirectories in shared datastore **130** corresponding to a particular tenant group corresponding may be synchronized such that a value written in subdirectory **222A** may be synchronized with subdirectory **222B** such that second shared service **124B** is operable to access values written by first shared service. In such embodiments, even if shared datastore **130** is operable to logically separate directories, rows, etc. accessible by various shared services **124** (e.g., a directory accessible via first shared service **124A** is not accessible via second shared service **124B**), certain information may be accessible to multiple shared service **124** in embodiments in which that certain information is synchronized.

(46) Accordingly, when tenants of multi-tenant cloud computing service **122** are owned by the same entity (e.g., ABC Corp.), and the tenant are members of the same tenant group, information in shared datastore **130** owned by ABC Corp. is able to be accessed by any of the members of the tenant group, acting through the various shared services **124**. If a first ABC Corp. tenant updates a value in shared datastore (e.g., the logistics department records a shipment arriving from a supplier being placed in inventory), other ABC Corp. tenants can access this value. For example, a second ABC Corp. tenant that is involved with maintaining the ABC Corp. website can access the updated inventory value and allow customers to purchase the item in question.

(47) FIG. 3 is a block diagram illustrating authentication and execution of a request made to shared service **124** in accordance with various embodiments. As depicted in FIG. 3, in various embodiments the authentication and execution of a request may be initiated by a remote computer system **110** (e.g., first remote computer system **110A**) communicating with a multi-tenant cloud computing service **122** (e.g., first multi-tenant cloud computing service **122A**) of a multi-tenant computer system **120** (e.g., through a website or API provided by first multi-tenant cloud computing service **122A**). As discussed herein, a tenant (e.g., tenant **A01 202**) corresponding to first remote computer system **110A** is provisioned at first multi-tenant cloud computing service **122A** (which itself is a tenant of multi-tenant computer system **120**), and requests to shared services **124** (e.g., first shared service **124A**) are made on behalf of this tenant corresponding to first remote computer system **110A**.

(48) In response to communication with first remote computer system **110A**, first multi-tenant cloud computing service **122A** sends a request to the corresponding shared service **124** (e.g., first shared service **124A**). In various embodiments, a first token **300** is sent with the request (or the request is included in a body portion of first token **300**). In various embodiments, first token **300** includes cryptographic information showing the authenticity of the request (e.g., tenant **A01 202** authorized the request) and payload information relating to the invocation of a tenant group (e.g., tenant group **A 200**) to perform the request. In various embodiments, the first token **300** includes an indication of tenant **A01 202** (e.g., a cryptographic indicator of tenant **A01 202** such as a public key or other value generated using a private key of tenant **A01 202**) and an indicator of tenant group **A 200** (e.g., a UUID or natural key corresponding to tenant group **A 200**). In various embodiments, first token **300** is a JavaScript Object Notation (JSON) web token (JWT) as defined in the RFC 7519 proposed standard published by the Internet Engineering Task Force (IETF) dated May 2015. An example first token **300** is shown in FIG. 5.

(49) In the embodiment shown in FIG. 3, first shared service **124A** receives a request from first

multi-tenant cloud computing service **122A** sent on behalf of tenant **A01202**. Subsequent to receiving a request, shared services **124** (e.g., first shared service **124A** as shown in FIG. 3) are operable to authenticate the request using cloud-to-cloud interface service **140** and tenant group module **150** and information in first token **300**. After authenticating the request, in various embodiments, the shared services **124** (e.g., first shared service **124A** as shown in FIG. 3) are operable to “wrap” first token **300** in a second token **302**, and send the second token **302** to shared datastore **130** as part of accessing shared datastore **130** to execute the request. In such embodiments, second token **302** contains information that can be verified to ensure that the request is indeed from a particular shared service **124**, as well as the contents of first token **300**, which identifies the tenant and tenant group. For example, in embodiments in which first token **300** is a JWT, second token **302** may be implemented as a chain of two JWTs (in which one of the JWTs in the chain is a copy of first token **300**). Shared datastore **130** is operable to authenticate the access using cloud-to-cloud interface service **140** and tenant group module **150** and information in in second token **302**. An example second token **302** is shown in FIG. 6.

(50) Cloud-to-cloud interface service **140** includes a cryptographic module **310** that is operable to perform the various cryptographic operations discussed herein. In various embodiments, cloud-to-cloud interface service **140** stores (or has access to) public keys for the various tenants of the multi-tenant computer services **122**. In various embodiments, the various multi-tenant computer services **122** generates respective private and public key pairs for their respective tenants. Then, the various multi-tenant computer services **122** send copies of the public keys to cloud-to-cloud interface service **140**. Then, when a shared service **124** receives a first token **300** claiming to be from a particular tenant of a particular multi-tenant computer service **122**, the shared service **124** is operable to authenticate first token **300** with cloud-to-cloud interface service **140**. For example, cryptographic module **310** is operable to authenticate a portion of first token **300** that is a cryptographic indicator of tenant **A01 202**. As discussed below, a portion of first token **300** (e.g., a signature section **520** shown in FIG. 5) is cryptographically signed using a private key corresponding to tenant **A01 202** or a signature generated using the private key and other portions of first token **300**. In various embodiments, cloud-to-cloud interface service **140** has a copy of the public key used by tenant **A01 202**. If cloud-to-cloud interface service **140** is able to validate a cryptographic signature in first token **300** with the copy of the public key on record for tenant **A01 202**, then first token **300** can be authenticated. In various embodiments, cloud-to-cloud interface service **140** also includes a tenant group identifier module **312** that is operable to generate the tenant grouping indicators **320** (e.g., tenant grouping indicator **320A**, tenant grouping indicator **320B**) discussed herein (e.g., the UUID and/or natural key for tenant group **A 200**).

(51) As discussed herein, tenant grouping module **150** is operable to authenticate requests by determining whether a requesting tenant is a member of the tenant group that is invoked in the request. In various embodiments, tenant grouping module **150** includes (or has access to) records of tenant grouping indicators **320** and tenant lists **322** for the various tenant groups. For example, tenant group module **150** includes tenant grouping indicators **320A** for tenant group **A 200** (e.g., the UUID and/or natural key for tenant group **A 200**) and tenant grouping indicators **320B** for tenant group **B 210** (e.g., the UUID and/or natural key for tenant group **B 200**). Additionally, tenant group module **150** includes tenant lists **322** for the various tenant groups such as tenant list **322A** for tenant group **A 200** and tenant list **322B** for tenant group **B 210**. As discussed herein, tenant lists **322** may include any of a number of indicators that identity the various tenants that are members of a tenant group (e.g., public keys of the tenants, UUIDs of the tenants). In various embodiments, to authenticate a request, shared services **124** and shared datastore **130** are operable to verify that a tenant is a member of the claimed tenant group by using tenant group module **150** to look up the claimed tenant group using a tenant grouping indicator included in the request (e.g., using a UUID corresponding to tenant group **A 200** included in first token **300**) and determine whether the tenant on whose behalf the request was sent appears on the tenant list **322** (e.g., determining that an

identifier of tenant **A01 202** is included on tenant list **322A**).

(52) Prior to using a tenant group to authenticate a request, the tenant group is provisioned by multi-tenant computer system **120** and one or more tenants are added to the tenant group. In various embodiments, a tenant of a multi-tenant cloud computing service **122** (e.g., tenant **A01 202**) can request that tenant group capability be enabled and cause a provisioning request to be sent from the multi-tenant cloud computing service **122** (e.g., to tenant group module **150**, to cloud-to-cloud interface service **140**). In response, a tenant group is created and one or more tenant group indicators are assigned (e.g., cloud-to-cloud interface service **140** assigned a UUID to the newly provisioned tenant group) and tenant group indicators are stored (e.g., a UUID and natural key corresponding to tenant group **A 200** is stored). In various embodiments, one or more tenants are identified as the owning tenant (e.g., the tenant that requested that the tenant group be created). After a tenant group has been created, one or more tenants may be added to the tenant group using subsequent provisioning requests. In various embodiments, these subsequent provisioning requests may be sent via the multi-tenant cloud computing service **122** of the owning tenant (e.g., because tenant **A01 202** is the primary member for tenant group **A 200**, subsequent provisioning requests may be sent via first multi-tenant cloud computer service **122A**). In various embodiments, subsequent provisioning requests are similar to first token **300** in that subsequent provisioning requests are signed using a private key of the owning tenant and include a tenant grouping indicator. The subsequent provisioning requests also include a payload with identifiers of the additional tenants to add to the tenant group. In various embodiments, only the owning tenant can request that additional tenants of the plurality of multi-tenant cloud computing services **122A** be added to the tenant group as members.

(53) In various embodiments, the various components of multi-tenant computer system **120** use a hierarchy of data protection scopes to determine whether requests are authorized. For example, a multi-tenant cloud computing service **122** may be operable to use data protection scopes to determine whether a particular tenant is authorized to send a request to a shared service. Similarly, shared services **124** may be operable to use data protection scopes to determine whether a particular tenant is authorized to send request to a particular shared service **124** and to determine whether a particular tenant is authorized to access shared datastore **130** in the context of the particular shared service **124**. FIG. 4 is block diagram of a data protection scope hierarchy **450** useable to determine whether requests are authorized in accordance with various embodiments. In various embodiments, data protection scope hierarchy **450** includes a plurality of levels (e.g., levels **400, 410, 420, 430, 440**) each of which includes one or more data protection scopes (e.g., data protection scopes **402** and **404** at level **400**, data protections scopes **412** at level **410**). As discussed herein, multi-tenant computer system **120** uses data protection scope hierarchy **450** to secure access to information (e.g., directories in shared datastore **130**) such that when a remote computer system **110** attempts to access secured information, multi-tenant computer system **120** consults data protection scope hierarchy **450** to determine whether to grant the access attempt. Data protection scope hierarchies are discussed in additional detail in U.S. patent application Ser. No. 16/588,550 entitled "Nested Tenancy That Permits A Hierarchy Having A Plurality Of Levels."

(54) In various embodiments, hierarchy **450** includes a plurality of levels. In the embodiment represented in FIG. 4, there are five levels: **400, 410, 420, 430, and 440**, although in various other embodiments there may be any of a number of levels numbering two or greater. In the embodiment shown in FIG. 4, level **400** is the top-level of the hierarchy **450** with levels beneath being subordinate to level **400**. For example, levels **410** and **420** include data protection scopes **412** and **422** that are directly subordinate to data protection scopes **402** and **404**, respectively. Thus, in various embodiments, levels **410** and **420** are subordinate to level **400**, level **430** is subordinate to level **420** (and in turn level **410**), and level **440** is subordinate to levels **400, 410, 420, and 430**.

(55) In data protection scope hierarchy **450**, subordinate data protection scopes (e.g., **412, 422, 424**) have access equal to or less than their superior data protection scopes (e.g., **402, 404**) in

various embodiments. In such embodiments, therefore, a tenant of a particular multi-tenant cloud computing service **122** would have access only to a subset of the information to which the particular multi-tenant computing service **122** has access. In various embodiments, data protection scopes at the same level are also isolated from each other such that at least some of the information accessible with a first data protection scope (e.g., First Cloud Computing Service Data Protection Scope **402A**) is isolated from at least some of the information accessible with a second data protection scope (e.g., Second Cloud Computing Service Data Protection Scope **402B**). In various embodiments, though, information is accessible by multiple data protection scopes. In some embodiments, such information that is made available to all data protection scopes at the same level of hierarchy **450**, or all data protection scopes of the same type (e.g., all cloud computing service data protection scope **402**). As discussed herein, an entity may use a tenant group to establish a trust relationship between various tenants it owns. In such instances, the data protection scopes corresponding to the grouped tenants may likewise be equalized. In various embodiments, for example, all members of a tenant group may be assigned to the data protection scope corresponding to the primary member of the tenant group.

(56) Level **400** is the top-most level of hierarchy **450**. Cloud computing service data protection scopes **402** and shared service data protection scopes **404** in the embodiment shown in FIG. **4**. In various embodiments, data protection scopes in level **400** are associated with first-class tenants of multi-tenant computer system. As discussed herein, levels in hierarchy **450** subordinate to level **400** are associated with subtenants of these first-class tenants or user accounts used to access tenants or subtenants.

(57) Cloud computing service data protection scope **402** are associated with multi-tenant cloud computing service **122** that have their own subtenants. In various embodiments in which multi-tenant computer system **120** is operated by SALESFORCE, examples of multi-tenant cloud computing service include a Customer Relationship Management (CRM) system that manages data for sales management for its subtenants, a Commerce Cloud that provides a unified commerce platform across multiple channels for subtenants, and a Marketing Cloud that provides digital marketing automation and analytics software and services for subtenants. As shown in FIG. **4**, two cloud computing service data protection scope **402** are provisioned in hierarchy **450**: First Cloud Computing Service Data Protection Scope **402A**, which has a first set of subtenants, and Second Cloud Computing Service Data Protection Scope **402B**, which has a second set of subtenants. While cloud computing service data protection scope **402** are shown in FIG. **4**, any number may be present in hierarchy **450**.

(58) Shared service data protection scopes **404** are associated with service products. As used herein, a shared service (a) “operates in the context of a multi-tenant cloud computing service **122**” when it is acting on behalf of a particular multi-tenant cloud computing service **122** and (b) “operates in the context of a tenant of a multi-tenant cloud computing service **122**” when it is acting on behalf of a particular tenant of a particular multi-tenant cloud computing service **122**. In various embodiments in which multi-tenant computer system **120** is operated by SALESFORCE, examples of shared services include an Inventory Service that is operable to manage an inventory of items (e.g., with holds on certain items, real-time availability, metrics) or an Asset Optimizer Service that is operable to manage metadata about digital assets for the tenant's commerce store. As shown in FIG. **4**, two shared service data protection scopes **404** are provisioned in hierarchy **450**: First Shared Service Data Protection Scope **404A** and Second Shared Service Data Protection Scope **404B**. As discussed below, subordinate data protection scopes are provisioned for shared service operating in the context of a multi-tenant cloud computing service **122** (e.g., First Shared Service Cloud Computing Service Data Protection Scopes **422**) or in the context of tenants of cloud products (e.g., first shared service cloud computing service tenant data protection scopes **432**). While two shared service data protection scopes **404** are shown in FIG. **4**, any number may be present in hierarchy **450**.

(59) Level **410** includes cloud computing service tenant data protection scopes **412** that are subordinate to cloud computing service data protection scopes **402** in level **400**. As shown in FIG. **4**, hierarchy **450** includes a plurality of First Cloud Computing Service Tenant Data Protection Scopes **412A** that are subordinate to First Cloud Computing Service Data Protection Scope **402A** and a plurality of Second Cloud Computing Service Tenant Data Protection Scopes **412B** that are subordinate to Second Cloud Computing Service Data Protection Scopes **402B**.

(60) Level **420** includes data protection scopes for shared services operating in the context of multi-tenant cloud computing services **122**: shared service cloud computing service data protection scopes **422**. These shared service cloud computing service data protection scopes **422** are subordinate to the data protection scopes for their respective service products, but these shared service cloud computing service data protection scopes **422** are also limited by the data scopes for their respective multi-tenant cloud computing services **122**. Thus, First Shared Service Cloud Computing Service Data Protection Scopes **422** are subordinate to First Shared Service Data Protection Scope **404A**, but because each of the First Shared Service Cloud Computing Service Data Protection Scopes **422** are operating in the context of different multi-tenant cloud computing services **122**, these scopes are also limited by the respective cloud computing service data protection scopes **402** (e.g., if first shared service **124A** is operating in the context of first multi-tenant cloud computing service **122A**, then the First Shared Service Cloud Computing Service Data Protection Scopes **422** is limited by First Cloud Computing Service Data Protection Scope **402A**).

(61) Level **430** includes data protection scopes for shared services operating in the context of tenants of multi-tenant cloud computing services **122**: shared service cloud computing service tenant data protection scopes **432** and shared service cloud computing service tenant data protection scopes **434**. These shared service cloud computing service tenant data protection scopes **432** are subordinate to the data protection scopes for their respective shared service products operating in the context of multi-tenant cloud computing services **122**, but these shared service cloud computing service data protection scopes **422** are also limited by (a) the data scopes for their respective multi-tenant cloud computing services **122** and (b) the data protection scopes for their respective tenants of those multi-tenant cloud computing services **122**. In some embodiments, shared service cloud computing service tenant data protection scopes **434** may be provisioned subordinate to shared service data protection scopes **404** (e.g., with no shared service cloud computing service data protection scopes).

(62) Level **440** includes user accounts **442** used by end users to access the data protection scopes in hierarchy **450**. In various embodiments, user accounts **442** associated with the same data protection scope have different levels of access to the information protected using hierarchy **450**. For example, a first user account **442** is associated with a salesman and a second user account **442** is associated with a sales manager who manages the salesman and other salesmen. In such embodiments, the second user account **442** has access to more of the protected data than the first user account **442**. However, user accounts **442** do not have access to more information than their respective data protection scopes, in various embodiments. In some embodiments, however, a single user account **442** may be associated with multiple data protection scopes, and may therefore have access to information beyond the scope of a single data protection scope. In some embodiments, user accounts **442** are only provisioned under cloud computing service data protection scopes **402**, but in other embodiments may optionally (as shown in the dashed lines) be provisioned under shared service data protection scopes **404** (e.g., in instances in which a user accounts may be used to access shared services **124** without using a multi-tenant cloud computing service).

(63) FIG. **5** is an example first token **300** sent from a multi-tenant cloud computing service **122** to a shared service **124** (e.g., a first token **300** sent from second multi-tenant cloud computing service **122B** to first shared service **124A**) in accordance with various embodiments. In the embodiment shown in FIG. **5**, first token **300** is formatted as a JWT with a header section **500**, a payload section

510, and a signature section **520**. In various instances, the header section **500** identifies first token **300** as a JWT and indicates the algorithm that was used to generate the signature section **520**. Any suitable cryptographic algorithm may be used including but not limited to HMAC with SHA-256 or RSA signature with SHA-256. The signature section **520** is usable to validate first token **300**. In various embodiments, signature section **520** is generated by encoding the contents of header section **500** and payload section **510** (e.g., by encoding the contents of header section **500** and payload section **510** using a binary-to-text encoding schema and concatenating the results), and then running the encoded results through the cryptographic algorithm identified in header section **500** and using a private key of the issuer of first token **300** (in this case, tenant **B01 204**).

(64) Payload section **510** includes an issuer (“iss”) entry **522** that identifies the principal that issued the JWT, an issuer type (“ist”) entry **524** that identifies the tenant group scope and instructs the recipient to treat first token **300** according to the tenant group authentication techniques discussed herein, an audience (“aud”) entry **526** that identifies the particular shared service **124** (i.e., first shared service **124A**) to which first token **300** is addressed, an audience type (“aut”) entry **528** that identifies the request to shared service **124** to be a request to a shared service, and a context (“ctx”) entry **530** that provides context for the request that is useable to ensure logical isolation (e.g., the context for the request is the second multi-tenant cloud computing service **122B** in the context of the multi-tenant computer system **120**). In the embodiment shown in FIG. 5, iss entry **522** includes an identifier **532** of the multi-tenant cloud computing service **122** from which the request was sent (i.e., second multi-tenant cloud computing service **122B**), an identifier **534** of the tenant of the multi-tenant cloud computing service **122** on whose behalf the request was sent (i.e., tenant **B01 204**), and an identifier **536** of the tenant group (i.e., a tenant grouping indication **320A** of tenant group **A 200**). In various embodiments, iss entry **522** also includes an indicator **538** of the primary member of the tenant group and an indicator of which multi-tenant cloud computing service **122** that primary member is a tenant (i.e., tenant **A01 202** of first multi-tenant cloud computing service **122A**).

(65) As discussed herein, a recipient of first token **300** is operable to authenticate the request for which first token **300** was sent (e.g., by authenticating the signature section **520**) and by verifying that the tenant identified in the iss entry **522** is a member of the tenant group identified in the iss entry **522**. Thus, in the example shown in FIG. 5, first shared service **124A** is able to authenticate the request by cryptographically authenticating signature section **520** with cloud-to-cloud interface service **140** and verifying membership of tenant **B01 204** in tenant group **A 200** with tenant group module **150**.

(66) FIG. 6 is an example second token **302** sent from a shared service **124** to a shared datastore **130** (e.g., a second token **302** sent from first shared service **124A** to shared datastore **130** as shown in FIG. 3) in accordance with various embodiments. In the embodiment shown in FIG. 6 (and similar to first token **300** shown in FIG. 5), second token **302** is formatted as a JWT with a header section **500**, a payload section **600**, and a signature section **610**. In various instances, the header section **500** identifies first token **300** as a JWT and indicates the algorithm that was used to generate the signature section **520**. Any suitable cryptographic algorithm may be used including but not limited to HMAC with SHA-256 or RSA signature with SHA-256. The signature section **610** is usable to validate second token **302**. In various embodiments, signature section **610** is generated by encoding the contents of header section **500** and payload section **600** (e.g., by encoding the contents of header section **500** and payload section **600** using a binary-to-text encoding schema and concatenating the results), and then running the encoded results through the cryptographic algorithm identified in header section **500** and using a private key shared service **124** (in this case, first shared service **124A**).

(67) Similar to payload section **510** of first token **300**, payload section **600** includes an issuer iss entry **522** that identifies the principal that issued the JWT. As shown in FIGS. 5 and 6, the iss entry **522** is the same for both first token **300** and second token **302**. In the embodiment shown in FIG. 6,

iss entry **522** includes an identifier **532** of the multi-tenant cloud computing service **122** from which the request was sent (i.e., second multi-tenant cloud computing service **122B**), an identifier **534** of the tenant of the multi-tenant cloud computing service **122** on whose behalf the request was sent (i.e., tenant **B01204**), and an identifier **536** of the tenant group (i.e., a tenant grouping indication **320A** of tenant group **A 200**). In various embodiments, iss entry **522** also includes an indicator **538** of the primary member of the tenant group and an indicator of which multi-tenant cloud computing service **122** that primary member is a tenant (i.e., tenant **A01 202** of first multi-tenant cloud computing service **122A**). Payload section **600** also includes an ist entry **602** that identifies the second token **302** as a proxy token (e.g., a token that is has been wrapped around another token another token from another issuer), an and entry **604** that identifies shared datastore **130** to which second token **302** is addressed, an aut entry **606** that identifies the request to shared datastore **130** to be targeted at the shared datastore **130** rather than any particular tenant of shared datastore **130**, and a ctx entry **608** that provides context for the request that is useable to ensure logical isolation (e.g., the context for the request is the shared datastore **130** in the context of first shared service **124A** in the context of the multi-tenant computer system **120**).

(68) As discussed herein, a recipient of second token **302** (e.g., shared datastore **130**) is operable to authenticate the request for which second token **302** was sent (e.g., by authenticating the signature section **610**) and by verifying that the tenant identified in the iss entry **522** is a member of the tenant group identified in the iss entry **522**. Thus, in the example shown in FIG. **6**, shared datastore **130** is able to authenticate the request by cryptographically authenticating signature section **610** with cloud-to-cloud interface service **140** and verifying membership of tenant **B01 204** in tenant group **A 200** with tenant group module **150**.

(69) Referring now to FIG. **7**, a flowchart depicting an authentication and execution method **700** is depicted. In the embodiment shown in FIG. **7**, the various actions associated with method **700** are implemented by multi-tenant computer system **120**. At block **702**, multi-tenant computer system **120** runs a shared service **124** (e.g., first shared service **124A**) operable to access a shared datastore **130** to execute requests from a plurality of multi-tenant cloud computing services **122** including a first multi-tenant cloud computing service **122A**. At block **704**, multi-tenant computer system **120** stores a first indication of a first tenant (e.g., tenant **A01 202**) of the first multi-tenant cloud computing service **122A** and a first tenant grouping indicator (e.g., a tenant grouping indicator **320A** for tenant group **A 200**), wherein the first tenant and one or more other tenants of the plurality of the multi-tenant cloud computing services **122** are associated with the first tenant grouping indicator. At block **706**, multi-tenant computer system **120** receives at the shared service **124** running on multi-tenant computer system **120**, from the first multi-tenant cloud computing service **122A**, a first request to the shared service **124**. The first request includes an indication of the first tenant and the first tenant grouping indicator. At block **708**, shared service **124** of the multi-tenant computer system **120** authenticates the first request by (a) cryptographically verifying the first request and (b) verifying that the first tenant is associated with the first tenant grouping indicator. At block **710**, in response to authenticating the first request, accessing, by the shared service running on multi-tenant computer system **120**, the shared datastore **130** to execute the first request.

(70) Exemplary Computer System

(71) Turning now to FIG. **8**, a block diagram of an exemplary computer system **800**, which may implement the various components of computer system **100** (e.g., multi-tenant computer system **120**) is depicted. Computer system **800** includes a processor subsystem **880** that is coupled to a system memory **820** and I/O interface(s) **840** via an interconnect **860** (e.g., a system bus). I/O interface(s) **840** is coupled to one or more I/O devices **850**. Computer system **800** may be any of various types of devices, including, but not limited to, a server system, personal computer system, desktop computer, laptop or notebook computer, mainframe computer system, tablet computer, handheld computer, workstation, network computer, a consumer device such as a mobile phone, music player, or personal data assistant (PDA). Although a single computer system **800** is shown in

FIG. 8 for convenience, system **800** may also be implemented as two or more computer systems operating together.

(72) Processor subsystem **880** may include one or more processors or processing units. In various embodiments of computer system **800**, multiple instances of processor subsystem **880** may be coupled to interconnect **860**. In various embodiments, processor subsystem **880** (or each processor unit within **880**) may contain a cache or other form of on-board memory.

(73) System memory **820** is usable to store program instructions executable by processor subsystem **880** to cause system **800** perform various operations described herein. System memory **820** may be implemented using different physical memory media, such as hard disk storage, floppy disk storage, removable disk storage, flash memory, random access memory (RAM-SRAM, EDO RAM, SDRAM, DDR SDRAM, RAMBUS RAM, etc.), read only memory (PROM, EEPROM, etc.), and so on. Memory in computer system **800** is not limited to primary storage such as memory **820**. Rather, computer system **800** may also include other forms of storage such as cache memory in processor subsystem **880** and secondary storage on I/O Devices **850** (e.g., a hard drive, storage array, etc.). In some embodiments, these other forms of storage may also store program instructions executable by processor subsystem **880**.

(74) I/O interfaces **840** may be any of various types of interfaces configured to couple to and communicate with other devices, according to various embodiments. In one embodiment, I/O interface **840** is a bridge chip (e.g., Southbridge) from a front-side to one or more back-side buses. I/O interfaces **840** may be coupled to one or more I/O devices **850** via one or more corresponding buses or other interfaces. Examples of I/O devices **850** include storage devices (hard drive, optical drive, removable flash drive, storage array, SAN, or their associated controller), network interface devices (e.g., to a local or wide-area network), or other devices (e.g., graphics, user interface devices, etc.). In one embodiment, computer system **800** is coupled to a network via a network interface device (e.g., configured to communicate over WiFi, Bluetooth, Ethernet, etc.), another example of an I/O device **850**.

(75) Although specific embodiments have been described above, these embodiments are not intended to limit the scope of the present disclosure, even where only a single embodiment is described with respect to a particular feature. Examples of features provided in the disclosure are intended to be illustrative rather than restrictive unless stated otherwise. The above description is intended to cover such alternatives, modifications, and equivalents as would be apparent to a person skilled in the art having the benefit of this disclosure.

(76) The scope of the present disclosure includes any feature or combination of features disclosed herein (either explicitly or implicitly), or any generalization thereof, whether or not it mitigates any or all of the problems addressed herein. Accordingly, new claims may be formulated during prosecution of this application (or an application claiming priority thereto) to any such combination of features. In particular, with reference to the appended claims, features from dependent claims may be combined with those of the independent claims and features from respective independent claims may be combined in any appropriate manner and not merely in the specific combinations enumerated in the appended claims.

Claims

1. A method of authenticating accesses to a shared datastore, the method comprising: running, at a computer system, a shared service operable to access the shared datastore to execute requests from a plurality of multi-tenant cloud computing services including a first multi-tenant cloud computing service; storing, at the computer system, a tenant grouping indicator that is indicative of a grouping of a first tenant of the first multi-tenant cloud computing service with one or more tenants of other ones of the plurality of multi-tenant cloud computing services that are permitted to access a set of resources managed by the shared service, wherein the first tenant is the only tenant associated with

the tenant grouping indicator that is allowed to request that additional tenants of the plurality of multi-tenant cloud computing services be associated with the tenant grouping indicator; receiving, at the computer system from the first multi-tenant cloud computing service, a first request to the shared service, wherein the first request includes a first token that includes a first indication of the first tenant and the tenant grouping indicator; in response to verifying that the first tenant is associated with the tenant grouping indicator, accessing, by the shared service, the shared datastore to execute the first request, wherein the accessing includes sending, to the shared datastore, a second token that includes an indication of the shared service and the first token; and wherein the second token is usable by the shared datastore to authenticate the first request, authenticate the accessing by the shared service, and verify that the first tenant is associated with the tenant grouping indicator.

2. The method of claim 1, wherein the first token includes an indication of a primary tenant associated with the tenant grouping indicator and an indication of the shared service, and wherein the second token includes an indication that the second token is a proxy token and targets the shared datastore.

3. The method of claim 1, further comprising: determining, by the computer system, that the first request is authorized using a hierarchy of data protection scopes that includes a shared service tenant group data protection scope that corresponds to the tenant grouping indicator.

4. The method of claim 1, further comprising: storing, at the computer system, a second indication of a second tenant of a second multi-tenant cloud computing service of the plurality of multi-tenant cloud computing services, wherein the second tenant is associated with the tenant grouping indicator; and based on authenticating a second request from the second multi-tenant cloud computing service, accessing, by the shared service running on the computer system, the shared datastore to execute the second request; wherein the first tenant and the second tenant correspond to the same entity; and wherein the accessing of the shared datastore to execute the first request includes writing a value to the shared datastore and the accessing of the shared datastore to execute the second request includes reading the value from the shared datastore.

5. The method of claim 1, further comprising: storing, at the computer system, a second indication of a second tenant of a second multi-tenant cloud computing service of the plurality of multi-tenant cloud computing services, wherein the second tenant is associated with the tenant grouping indicator; and wherein tenancy in the first multi-tenant cloud computing service is defined by a first set of parameters and tenancy in the second multi-tenant cloud computing service is defined by a second set of parameters.

6. The method of claim 1, wherein the accessing of the shared datastore to execute the first request includes accessing a particular directory of the shared datastore corresponding to the shared service and the tenant group indicator and updating a value stored in the particular directory, wherein the particular directory is logically isolated from other directories corresponding to other tenant group indicators.

7. The method of claim 1, further comprising: running, at the computer system, a second shared service operable to: access the shared datastore to execute requests from the plurality of multi-tenant cloud computing services; and authenticate requests made to the second shared service from the plurality of multi-tenant cloud computing systems using the tenant grouping indicator; wherein tenant grouping is the only representation of trust relationships between tenants of the plurality of multi-tenant cloud computing services used by the shared service and the second shared service to authenticate requests.

8. A non-transitory computer-readable storage medium storing program instructions that are capable of being executed by a computer system to perform operations comprising: running a shared service operable to access a shared datastore to execute requests from a plurality of multi-tenant cloud computing services including a first multi-tenant cloud computing service; storing a tenant grouping indicator that is indicative of a grouping of a first tenant of the first multi-tenant

cloud computing service with one or more tenants of other ones of the plurality of multi-tenant cloud computing services that are permitted to access a set of resources managed by the shared service; receiving, from the first multi-tenant cloud computing service, a first request to the shared service, wherein the first request includes a first token that includes an indication of the first tenant and the tenant grouping indicator; determining that the first request is authorized using a hierarchy of data protection scopes that includes a shared service tenant group data protection scope that corresponds to the tenant grouping indicator; in response to verifying that the first tenant is associated with the tenant grouping indicator, accessing the shared datastore to execute the first request, wherein the accessing includes sending, to the shared datastore, a second token that includes an indication of the shared service and the first token; and wherein the second token is usable by the shared datastore to authenticate the first request, authenticate the accessing by the shared service, and verify that the first tenant is associated with the tenant grouping indicator.

9. The non-transitory computer-readable storage medium of claim 8, wherein the first token includes an indication of a primary tenant associated with the tenant grouping indicator and an indication of the shared service, and wherein the second token includes an indication that the second token is a proxy token and targets the shared datastore.

10. The non-transitory computer-readable storage medium of claim 8, wherein the first tenant is the only tenant associated with the tenant grouping indicator that is allowed to request that additional tenants of the plurality of multi-tenant cloud computing services be associated with the tenant grouping indicator.

11. The non-transitory computer-readable storage medium of claim 8, wherein the operations further comprise: storing a second indication of a second tenant of a second multi-tenant cloud computing service of the plurality of multi-tenant cloud computing services, wherein the second tenant is associated with the tenant grouping indicator; and based on authenticating a second request from the second multi-tenant cloud computing service, accessing the shared datastore to execute the second request; wherein the first tenant and the second tenant correspond to the same entity; and wherein the accessing of the shared datastore to execute the first request includes writing a value to the shared datastore and the accessing of the shared datastore to execute the second request includes reading the value from the shared datastore.

12. The non-transitory computer-readable storage medium of claim 8, wherein the operations further comprise: storing a second indication of a second tenant of a second multi-tenant cloud computing service of the plurality of multi-tenant cloud computing services, wherein the second tenant is associated with the tenant grouping indicator; and wherein tenancy in the first multi-tenant cloud computing service is defined by a first set of parameters and tenancy in the second multi-tenant cloud computing service is defined by second set of parameters.

13. The non-transitory computer-readable storage medium of claim 8, wherein the operations further comprise: running a second shared service operable to access the shared datastore to execute requests from the plurality of multi-tenant cloud computing services, wherein the second shared service is operable to authenticate requests made to the second shared service from the plurality of multi-tenant cloud computing services using the tenant grouping indicator; wherein tenant grouping is the only representation of trust relationships between tenants of the plurality of multi-tenant cloud computing services used by the shared service and the second shared service to authenticate requests.

14. A system, comprising: at least one processor; and memory having program instructions stored thereon that are executable by the at least one processor to perform operations comprising: running a shared service operable to access a shared datastore to execute requests from a plurality of multi-tenant cloud computing services including a first multi-tenant cloud computing service; storing a tenant grouping indicator that is indicative of a grouping of a first tenant of the first multi-tenant cloud computing service with one or more tenants of other ones of the plurality of multi-tenant cloud computing services that are permitted to access a set of resources managed by the shared

service; receiving, from the first multi-tenant cloud computing service, a first request to the shared service, wherein the first request includes a first token that includes an indication of the first tenant and the tenant grouping indicator; in response to verifying that the first tenant is associated with the tenant grouping indicator, accessing the shared datastore to execute the first request, wherein the accessing includes sending, to the shared datastore, a second token that includes an indication of the shared service and the first token; and wherein the second token is usable by the shared datastore to authenticate the first request, authenticate the accessing by the shared service, and verify that the first tenant is associated with the tenant grouping indicator, and wherein the accessing of the shared datastore to execute the first request includes accessing a particular directory of the shared datastore corresponding to the shared service and the tenant group indicator and updating a value stored in the particular directory, wherein the particular directory is logically isolated from other directories corresponding to other tenant group indicators.

15. The system of claim 14, wherein the operations further comprise: running a second shared service operable to access the shared datastore to execute requests from the plurality of multi-tenant cloud computing services, wherein the second shared service is operable to authenticate requests made to the second shared service from the plurality of multi-tenant cloud computing services using the tenant grouping indicator; wherein tenant grouping is the only representation of trust relationships between tenants of the plurality of multi-tenant cloud computing services used by the shared service and the second shared service to authenticate requests.

16. The system of claim 14, wherein the first tenant is the only tenant associated with the tenant grouping indicator that is allowed to request that additional tenants of the plurality of multi-tenant cloud computing services be associated with the tenant grouping indicator.

17. The system of claim 14, wherein the operations further comprise: determining, using a hierarchy of data protection scopes including a shared service tenant group data protection scope that corresponds to the tenant grouping indicator, that the first request is authorized.
